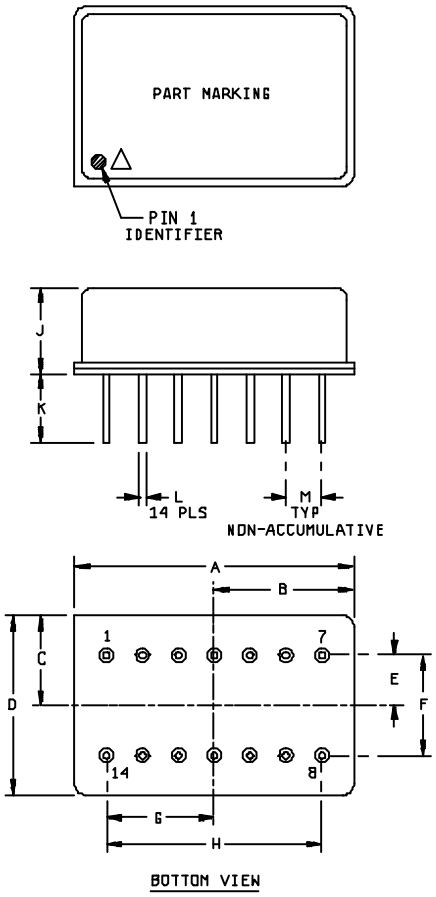
INCH-POUND
MIL-PRF-55310/25D
11 June 2009
SUPERSEDING
MIL-PRF-55310/25C
7 October 2004

PERFORMANCE SPECIFICATION SHEET

OSCILLATOR, CRYSTAL CONTROLLED, TYPE 1 (CRYSTAL OSCILLATOR (XO)), 25 MHz THROUGH 175 MHz, HERMETIC SEAL, SQUARE WAVE, EMITTER COUPLED LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-55310.



Pin number	Function
1	NC
2	NC
3	NC
4	NC
5	NC
6	NC
7	B-
8	OUTPUT
9	NC
10	NC
11	NC
12	NC
13	NC
14	GND/CASE

Configuration A

FIGURE 1. Dimensions and configuration.

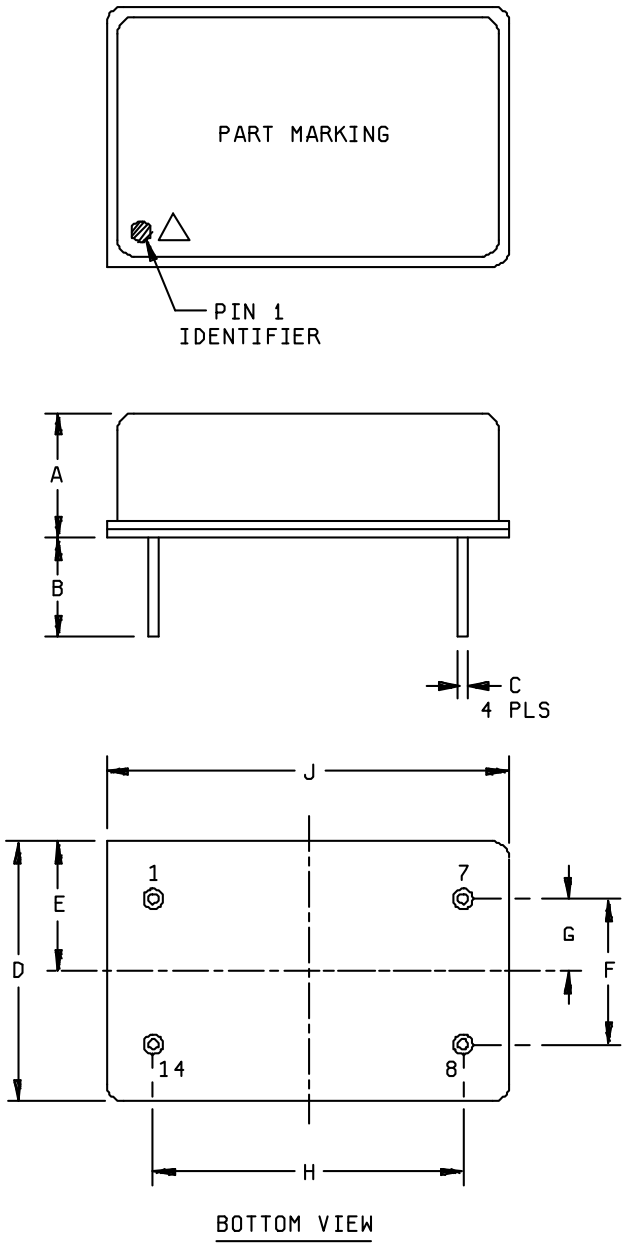
Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.887	---	22.53
B	---	.44	---	11.2
C	---	.27	---	6.8
D	---	.54	---	13.7
E	.145	.155	3.68	3.94
F	.295	.305	7.49	7.75
G	.295	.305	7.49	7.75
H	.595	.605	5.11	5.37
J	---	.20	---	5.1
K	.20	---	5.1	---
L	.016	.020	0.41	0.51
M	.095	.105	2.41	2.67

Configuration A

FIGURE 1. Dimensions and configuration - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.



Pin number	Function
1	NC
7	B-
8	Output
14	GND/case

Configuration B

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.20	---	5.1
B	.20	---	5.1	---
C	.016	.020	0.41	0.51
D	---	.515	---	13.08
E	---	.27	---	6.8
F	.295	.305	7.49	7.75
G	.145	.155	3.68	3.94
H	.595	.605	15.11	15.37
J	---	.815	---	20.70

Configuration B

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) for three place decimals and ±.02 (0.5 mm) for two place decimals.
4. All pins with the NC function may be connected internally and are not to be used as external tie points or connections.
5. Color dot or square corner shall be used to indicate pin number 1.
6. ESD indicator, open triangle, shall be marked anywhere on the top of the oscillator.

FIGURE 1. Dimensions and configurations - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Interface and physical dimensions: See figure 1.

Package configuration: See figure 1.

Terminals: See figure 1.

Weight: 0.5 ounces (14.175 grams), maximum.

Oscillator: Class 2 or any class 1 or class 3 oscillator meeting all class 2 requirements and verification tests specified herein and in MIL-PRF-55310.

Calibration: Manufacturer calibrated.

Screening: In accordance with MIL-PRF-55310, product level B or S, as applicable.

Temperature:

Operating: See table I.

Storage: -62°C to +125°C.

Load test circuit: See figure 2.

Seal: Hermetic in accordance with MIL-PRF-55310, maximum leakage rate 5 x 10^{-8} atm cc/s.

Supply voltage (B-): -5.2 V dc ±5 percent.

Overvoltage survivability: In accordance with MIL-PRF-55310.

Supply current: At designated supply voltage (see table I).

Start-up time: 15 milliseconds maximum, measurement shall be taken at reference temperature and operating temperature range endpoints.

Nominal output frequency: Frequency as designated at time of acquisition (see table I).

Output logic voltage levels: See figure 3.

Logic 1: -1.15 V dc minimum.

Logic 0: -1.54 V dc maximum.

Output waveform: Symmetrical square wave, emitter coupled logic compatible (see figure 3).

Duty cycle: See table I and figure 3.

Rise and fall times (see table I): Measurements shall be taken at the 20 percent and 80 percent peak-to-peak output voltage levels, with peak-to-peak output defined as Logic 1 - Logic 0 (see figure 3).

Initial accuracy at reference temperature: See table I.

Initial frequency-temperature accuracy (1/2 temperature cycle): See table I. Measurements shall be taken at ten equally spaced increments over the specified operating temperature range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

TABLE I. Dash numbers and operating characteristics.

Dash number		Output frequency range	Input current max at 5.25 V ±5% 2/	Pulse characteristics			Initial accuracy at +23°C ±1°C 3/	Frequency aging per year (max) 4/	Frequency-temperature tolerance (ppm)		
Config-uration				Rise and fall times max 1/	Duty cycle min-max 1/	Load (test) to -2.0 V dc 2/			-55°C to +125°C	-55°C to +105°C	-20°C to +70°C
A	B								A	B	C
			mA	ns	percent	ohms	ppm	ppm	ppm	ppm	ppm
02	03	25 MHz to 100 MHz	50	3.5	40 to 60	50	±15	±5	±65	±55	±40
06	07	25 MHz to 100 MHz	75	3.5	40 to 60	100	±15	±5	±65	±55	±40
10	11	25 MHz to 100 MHz	50	3.5	40 to 60	50	±25	±10	±100	±75	±50
14	15	25 MHz to 100 MHz	75	3.5	40 to 60	100	±25	±10	±100	±75	±50
32	33	100 MHz to 125 MHz	60	3.5	40 to 60	50	±15	±5	±65	±55	±40
36	37	100 MHz to 125 MHz	75	3.5	40 to 60	100	±15	±5	±65	±55	±40
40	41	100 MHz to 125 MHz	60	3.5	40 to 60	50	±25	±10	±100	±75	±50
44	45	100 MHz to 125 MHz	60	3.5	40 to 60	100	±25	±10	±100	±75	±50
62	63	125 MHz to 175 MHz	70	3.0	40 to 60	50	±15	±5	±65	±55	±40
66	67	125 MHz to 175 MHz	125	3.0	40 to 60	100	±15	±5	±65	±55	±40
70	71	125 MHz to 175 MHz	70	3.0	40 to 60	50	±25	±10	±100	±75	±50
74	75	125 MHz to 175 MHz	125	3.0	40 to 60	100	±25	±10	±100	±75	±50

1/ See figure 3.
2/ See figure 2.
3/ Up to 30 days following shipment.
4/ After 30 days following shipment.

TABLE II. Frequency aging limits.

	5 ppm per year 1/	10 ppm per year 1/
Maximum change over 30 days	±0.7 ppm	±1.5 ppm
Projected maximum change for 1 year after 30 days	±5.0 ppm	±10.0 ppm

1/ See table I.

Frequency-voltage tolerance ±1 ppm maximum for a ±5 percent change in supply voltage. Measurements shall be taken at reference temperature and operating temperature range endpoints.

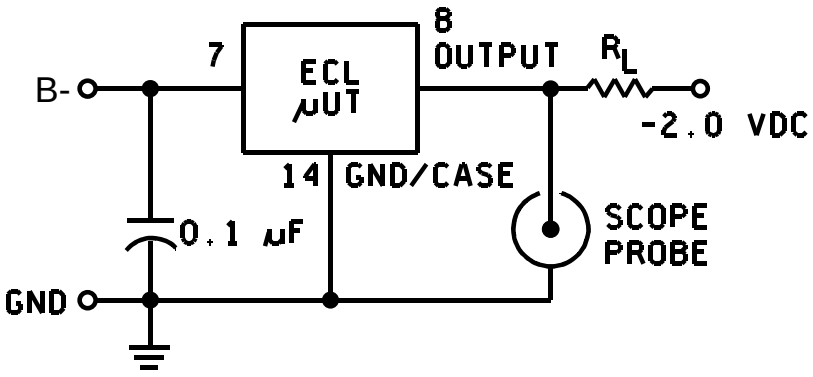
Frequency-environmental tolerance: ±3 ppm.

Frequency aging: Frequency aging shall be in accordance with MIL-PRF-55310 and shall meet the limits of table II.

Vibration, sinusoidal: Method 204 of MIL-STD-202.

Nonoperating: Test condition D.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.



NOTE: R_L = 50Ω or 100Ω (see table I).

FIGURE 2. Load test circuit.

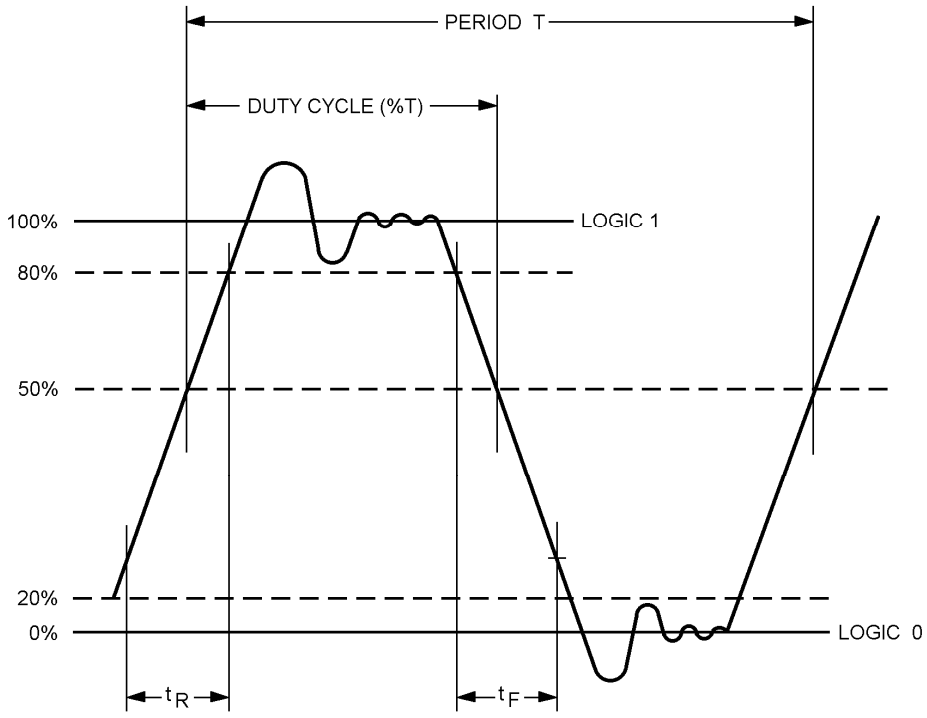


FIGURE 3. Waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

Operating: Not required.

Shock (nonoperating): Method 213 of MIL-STD-202, test condition I.

Thermal shock (nonoperating): Method 107 of MIL-STD-202, test condition B.

Ambient pressure:

Nonoperating: In accordance with MIL-PRF-55310.

Operating: Method 105 of MIL-STD-202, test condition C.

Resistance to soldering heat: Method 210 of MIL-STD-202, test condition E.

Moisture resistance: Method 106 of MIL-STD-202.

Terminal strength: Method 211 of MIL-STD-202, condition C.

Applied force: 2 pounds each terminal for 10 seconds.

Bends: 5 at 45 degrees each.

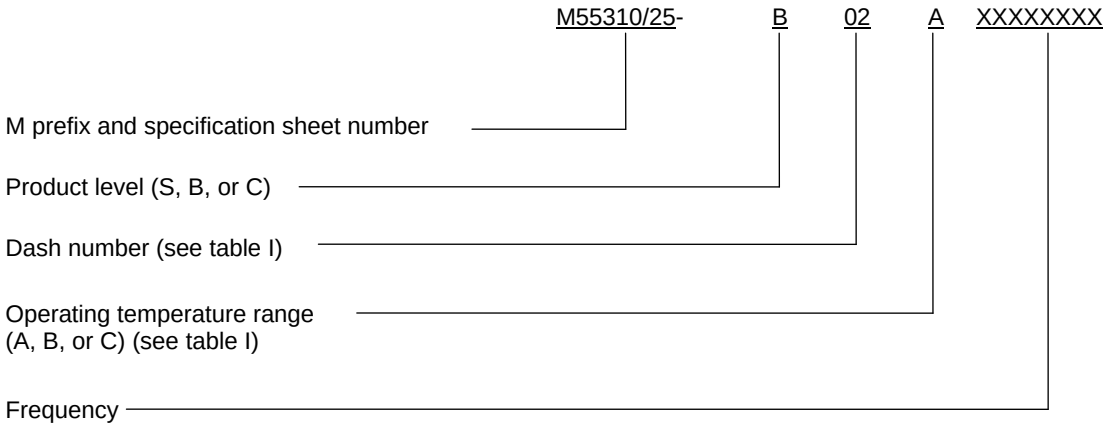
Solderability: Method 208 of MIL-STD-202.

Resistance to solvents: Method 215 of MIL-STD-202.

Screening: In accordance with MIL-PRF-55310, class B or S, as applicable.

Part or Identifying Number (PIN): Consists of “M” prefix followed by specification sheet number, a dash and coded alphas, and numeric number. See example:

EXAMPLE



Reference documents. In addition to MIL-PRF-55310, this document references the following:

MIL-STD-202

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 84
NASA - NA

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-2009-020)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T23:00Z
Check the source to verify that this is the current version before use.